

RELEASE-DAY BASELINE DIAGNOSTIC W1-W4

Full name: Lê Nhữ Nghĩa

Student ID: 2453402

Part 1 - ML Foundations

1a) State Mitchell's definition. Identify T, P, E for recognizing human activities from smartphone sensors

Mitchell's definition: "A program learns from experience E with respect to task T & performance P if its performance at T , measured by P , improves with E "

T : classify a window of accelerometer & gyroscope signals into G activities $\Rightarrow$ Supervised classification
 E : labelled tensor windows recorded from volunteers
 P : accuracy, precision, recall, F1 & macro-F1 on unseen data

1b) Distinguish overfitting & underfitting. How does each show up in training error & validation error?

	Underfitting (high bias)	Overfitting (high variance)
Cause	Model too simple	Model too complex, memorizes noise
Train error	High	Low
Val / test error	High, close to train	Much higher than train (large gap)
Remedy	More complex, more features	Regularisation, more data, pruning

1c) Write the bias-variance decomposition of the expected squared error at a point u . Briefly explain each term. Which term cannot be reduced?

With $y = f(x) + \epsilon$, $E[\epsilon] = 0$, $\text{Var}(\epsilon) = \sigma^2$

$$E[(y - \hat{f}(u))^2] = (E[\hat{f}(u)] - f(u))^2 + E[(\hat{f}(u) - E[\hat{f}(u)])^2] + \sigma^2$$

= Bias² + Variance + Noise
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 Model too simple to represent f (underfitting) Sensitivity to the training sample (overfitting) Irreducible; no model can remove it

1d) Given the 3-class confusion matrix rows = true, columns = predict

True \ Predicted	A	B	C
A	8	2	0
B	1	6	3
C	0	0	5

Compute: accuracy, per-class precision/recall/F1, macro-F1

True: $A=10$ Predicted: $A=9$ $N=25$
 $B=10$ $B=8$ Diagonal = 19 } $\rightarrow \text{Accuracy} = \frac{19}{25} = 0.76$
 $C=5$ $C=8$

Class	C_{kk}	Precision	Recall	F1
A	8	8/9	8/10 = 0.8	0.842
B	6	6/8	6/10 = 0.6	0.667
C	5	5/8	5/5 = 1	0.769

$$\text{Macro-F1} = \frac{0.842 + 0.667 + 0.769}{3} = 0.759$$

$$\text{Weighted F1} = 0.757 ; \text{micro-F1} = \text{accuracy} = 0.76$$

Part 2 - Decision Tree

2a) write formulas for Entropy, Information Gain, & Gini impurity. How does IG favour attributes with many values? Which quantity does C4.5 use to correct this?

$$\text{Entropy: } H(S) = -\sum p_i \log p_i ; \text{IG}(S, A) = H(S) - \sum_v \frac{|S_v|}{|S|} \cdot H(S_v) ; \text{Gini} = 1 - \sum p_i^2$$

IG biases an attribute with many values with split S into tiny, nearly pure datasets

$\hookrightarrow \text{IG} \approx H(S)$ although the split does not generalise (like an ID column)

C4.5 fix with Gain Ratio:

$$\text{Split Info}(S, A) = -\sum_v \frac{|S_v|}{|S|} \log_2 \left(\frac{|S_v|}{|S|} \right) ; \text{GR} = \frac{\text{IG}}{\text{Split Info}}$$

2b) Two ways to handle missing attribute values, both when building trees & when classifying a new sample

Naïve options: drop rows (wasteful) or impute mean/mode (may bias). Tree-specific methods:

- C4.5 Fractional weighting:

- Build: } compute IG/GR on samples where A is observed
- } send samples missing A to every child with weights proportional to branch frequencies
- Classify: } send the sample down all branches with those weights
- } combine leaf distributions, predict heaviest class

- CART surrogate splits:

- Build: } pick the split using observed A
- } store top- k other features whose splits best agree with it

- Classify: if A is missing, route with the best available surrogate, then the next one

2c) Distinguish pre-pruning & post-pruning. For cost-complexity pruning $R_\alpha(T) = R(T) + \alpha|T|$: what do the symbols mean, how does the tree change as α increases, & how is α chosen?

- Pre-pruning: stop growing when $|S| < N_{\min}$, the gain is below threshold, or the node is nearly pure & the node becomes a majority leaf (early stopping)

- Post-pruning (C4.5): grow fully then bottom-up replace a subtree by a leaf if it's not significantly better (pessimistic error $\hat{p} + 2\sqrt{\hat{p}(1-\hat{p})/N}$)

$$R_\alpha(T) = R(T) + \alpha|T|$$

$\downarrow$ Training Error $\downarrow$ Number of leaves

($\alpha = 0$ gives full tree; larger α gives smaller tree)

⑦ Weakest-link pruning: repeatedly removes the node with the smallest $g(t) = \frac{R(t) - R(T_f)}{|T_t| - 1}$, giving a nested sequence of subtrees

Choose α by cross-validation (optimally the one-SE rule)